E7: Introduction to Programming & Numerical Methods
Midterm 2A

Tuesday, November 04

Full Name

Student ID Number

Instructions:

- You have 1 hour and 20 minutes to complete this exam.
- The exam is closed book, closed notes, closed computer, closed calculator.
- You may only use one 8.5" × 11" sheet of notes (both sides). No other resources are permitted.
- Mark your answers on the exam paper within the designated spaces. Responses written on scratch paper or outside of the specified answer spaces will not be graded.
- This exam is DOUBLE-SIDED. Make sure to read both sides of each page.
- The number of points associated with each part of each question is indicated.
- The exam is worth a total of 100 points.
- For questions with circular bubbles, you should select exactly *one* choice.
  - ◯ You may only select one option!
  - ◯ Or this one, but not both!
  - ◯ Or this one, but no more than one!
- For questions with square boxes, you may select *multiple* choices.
  - ☐ You can select this one!
  - ☐ You can select this one too!
  - ☐ You can select one, two, or all three of these!
- Only include your FINAL answers in the designated boxes. Any additional text will count as part of your answer and may result in a reduced grade.

Good luck!

Question 1 (30 points)

There are 11 multiple choice questions. There is only **ONE** correct choice for each question. Multiple bubbles, incomplete bubbles, or stray marks will be marked incorrect.

1. (3 pt) Which of the following statements about number representation in computers is False?
   - ◯ There is a finite number of representable numbers
   - ◯ There is a non-zero gap between consecutive representable numbers
   - ● None of the statements are False
   - ◯ There is a minimum representable number
   - ◯ There is a maximum representable number

2. (3 pt) Assume that you defined two matrices `A` and `B` in MATLAB and then obtained the output below. Which of the following MATLAB expressions will NOT result in an error?

```
>> size(A)
ans =
     3   3

>> size(B)
ans =
     4   3
```

   - ◯ `B .* A`
   - ◯ `A .* B`
   - ◯ `A .^ B`
   - ● `B * A`
   - ◯ `A * B`
   - ◯ `A + B`
   - ◯ None of the options result in an error

3. (3 pt) Which of the following correctly uses an `if` statement in MATLAB?
   - ◯
```
if x > 10:
    x = x + 2;
end
```
   - ●
```
if x > 10
x = x + 2
end
```
   - ◯
```
if x > 10:
x = x + 2
```
   - ◯
```
if x > 10
x = x + 2
```
   - ◯
```
if x > 10:
    x = x + 2
```
   - ◯
```
if x > 10:
x = x + 2
end
```

4. (3 pt) What is the output of the following MATLAB code?

```
nums = [2, 6, 4, 8, 10];
disp(nums(2))
```

   - ◯ 8
   - ◯ 4
   - ● 6
   - ◯ 2
   - ◯ Error
   - ◯ 10

5. (1 pt) At what time does our E7 lecture start?
   - ◯ 2:00 PM
   - ◯ 7:00 PM
   - ◯ 6:00 PM
   - ● 5:00 PM
   - ◯ 3:30 PM
   - ◯ 6:30 PM

6. (3 pt) Which of the following is NOT a valid MATLAB operator?
   - ◯ `<=`
   - ● `!=`
   - ◯ `>=`
   - ◯ `==`
   - ◯ `||`
   - ◯ `&`
   - ◯ `*`

7. (2 pt) What will be displayed when you run the following MATLAB code?

```
one = 1;
two = 2;
if one < two
disp(1)
elseif one + one == two
disp(2)
if one < two
disp(3)
else
disp(4)
end
end
```

- ○ 1 2
- ○ 1 3
- ● 1
- ○ 1 4
- ○ 3
- ○ 1 2 3
- ○ 4
- ○ 2

8. (3 pt) What is the value of `v` after executing the following line in MATLAB: `v = 10 % 2`

○ 1 ○ Error ○ 2 ○ False ○ 5 ○ 0 ● 10

9. (3 pt) Which of the following statements is <u>False</u> after executing the following code?

```
import numpy as np
x = y = np.linspace(-10, 10.5, 10)
X, Y = np.meshgrid(x, y)
```

- ● `X == Y` evaluates to an array of `True` values
- ○ The code runs without raising any error
- ○ `X` has 100 elements
- ○ `x == y` evaluates to an array of `True` values
- ○ Every element of `x` is equal to at least one element of `Y`
- ○ Every element of `x` is equal to at least one element of `X`

10. (3 pt) Which of the following statements about `try-except` blocks is <u>True</u>?

- ● The `try` block should contain code that you expect may cause an exception
- ○ They can handle any type of error
- ○ They can only handle one type of exception at a time
- ○ Specifying the error type after `except` is required
- ○ Using `try-except` will always prevent a code from crashing

11. (3 pt) Consider a file named `data.txt` located in the same directory as your code. Which option writes a single line `200 210 220` to the file without deleting its existing content?

○
```
with open('data.txt', 'a') as f:
    f.write(200 210 220)
```

●
```
with open('data.txt', 'a') as f:
    f.write('200 210 220')
```

○
```
with open('data.txt', 'r') as f:
    f.write('200 210 220')
```

○
```
with open('data.txt', 'w') as f:
    f.write(200 210 220)
```

○
```
with open('data.txt', 'w') as f:
    f.write('200 210 220')
```

○
```
with open('data.txt', 'a') as f:
    f.write('200, 210, 220')
```

Question 2 (10 points)

The following might be helpful for some of the next parts:

$2^0 = 1, 2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16.$ $4^0 = 1, 4^1 = 4, 4^2 = 16, 4^3 = 64, 4^4 = 256.$

$3^0 = 1, 3^1 = 3, 3^2 = 9, 3^3 = 27, 3^4 = 81.$ $5^0 = 1, 5^1 = 5, 5^2 = 25, 5^3 = 125, 5^4 = 625.$

(a) (2 pt) A sensor reports acceleration using a 6-bit signed binary integer. What is the maximum positive value that can be represented? Provide your answer as a decimal (base-10) integer. You must show your work and write your final answer below.

Your work:

$2^4 + 2^3 + 2^2 + 2^1 + 2^0 = 31$

Your final answer:

31

(b) (3 pt) The velocity is stored using a fixed-size integer type with a maximum representable decimal value of 511. Let `V0` represent the initial velocity, and `V1` represent the velocity after 1 minute. What happens when the operation below is performed? Select ALL that apply.

```
V0 = 510
V1 = V0 + 5
```

- ☐ The result is clipped to the maximum representable value
- ☐ `V1` correctly stores 515
- ☐ The operation converts the integer to floating-point automatically
- ■ The operation may produce an unexpected number or an error message
- ☐ An underflow occurs
- ☐ There is a round-off error
- ■ An overflow occurs

(c) (3 pt) Convert the following unsigned quaternary (base-4) integer to its decimal equivalent. You must show your work and write your final answer below. Quaternary number: `123`

Your work:

$1 * 4^2 + 2 * 4^1 + 3 * 4^0 = 27$

Your final answer:

27

(d) (2 pt) In a floating-point number (single or double precision), increasing the exponent by 1:

- ● Doubles the represented number
- ○ Increases the represented number by 1
- ○ Divides the represented number by 2
- ○ Increases the represented number by 2
- ○ Changes the exponent to a negative value
- ○ None of the options
- ○ Leaves the represented number unchanged
- ○ Results in round-off error

Question 3 (10 points)

Consider the following arrays in Python. Your task is to write their MATLAB equivalents with exactly the same values. **For parts (a) and (c), you may not use any functions.**

```
import numpy as np
R = np.arange(4, 22, 0.1)   # [4.0, 4.1, 4.2, 4.3, 4.4, ..., 21.7, 21.8, 21.9]
B = np.linspace(5, 25, 26)  # [5.0, 5.8, 6.6, 7.4, 8.2, ..., 23.4, 24.2, 25.0]
V = np.arange(180)          # [0, 1, 2, 3, 4, ..., 177, 178, 179]
D = np.cos(np.pi/9)
```

(a) (3 pt) Write a one-line MATLAB command that correctly reproduces vector `R`.

```
R = 4:0.1:21.9
```

(b) (2 pt) Which of the following MATLAB expressions correctly reproduces vector `B`?

- ● `linspace(5, 25, 26)`
- ○ `5:25:25`
- ○ `np.linspace(5, 26, 25)`
- ○ `linspace(5, 26, 25)`
- ○ `linspace(5, 25, 25)`
- ○ `np.linspace(5, 25, 25)`
- ○ `np.linspace(5, 25, 26)`
- ○ `5:26:25`
- ○ `5:25:26`

(c) (2 pt) Write a one-line MATLAB command that correctly reproduces vector `V`.

```
V = 0:179 % OR 0:1:179
```

(d) (1 pt) Write a one-line MATLAB command that correctly reproduces the value `D`.

```
D = cos(pi/9)
```

(e) (2 pt) Assume vectors `R` and `V` have been defined and represent the resistances and voltages of a series of resistors. Both vectors have the same number of elements. Compute a new vector `P`, where each element represents the power dissipated by a resistor, calculated as the square of its voltage divided by its resistance. Which of the following expressions computes `P` in MATLAB?

- ○ `P = V.**2 / R`
- ○ `P = R.**2 ./ V`
- ○ `P = R**2 / V`
- ○ `P = V.^2 / R`
- ○ `P = V.**2 ./ R`
- ○ `P = R^2 ./ V`
- ○ `P = R**2 ./ V`
- ○ `P = V^2 ./ R`
- ○ `P = V**2 / R`
- ○ `P = R.^2 ./ V`
- ○ `P = R^2 / V`
- ● `P = V.^2 ./ R`
- ○ `P = R.^2 / V`
- ○ `P = R.**2 / V`
- ○ `P = V^2 / R`
- ○ `P = V**2 ./ R`

Question 4 (15 points)

Three metal plates are connected and exchange heat. Let T_1, T_2, T_3 be the temperatures of plates 1, 2, and 3, respectively. Using steady-state energy balances, you developed the following system of linear equations in matrix form $\mathbf{MT} = \mathbf{y}$:

$$\begin{cases} 3T_1 - T_2 - T_3 = 50 \\ 4T_2 - T_1 - T_3 = 100 \\ T_1 + T_2 + T_3 = 120 \\ 2T_2 + 3T_3 = 350 \end{cases} \qquad \mathbf{MT} = \mathbf{y} \rightarrow \mathbf{M} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} = \mathbf{y}$$

(a) (2 pt) Which of the following is the correct representation of matrix $\mathbf{M}$?

○ $\begin{bmatrix} 3 & -1 & -1 \\ -1 & 4 & -1 \\ 1 & 1 & 1 \\ 2 & 3 & 0 \end{bmatrix}$ ○ $\begin{bmatrix} 3 & -1 & -1 & 50 \\ 4 & -1 & -1 & 100 \\ 1 & 1 & 1 & 120 \\ 0 & 2 & 3 & 350 \end{bmatrix}$ ○ $\begin{bmatrix} 3 & -1 & -1 & 50 \\ 4 & -1 & -1 & 100 \\ 1 & 1 & 1 & 120 \\ 2 & 3 & 0 & 350 \end{bmatrix}$ ○ $\begin{bmatrix} 3 & -1 & -1 \\ 4 & -1 & -1 \\ 1 & 1 & 1 \\ 2 & 3 & 0 \end{bmatrix}$

● $\begin{bmatrix} 3 & -1 & -1 \\ -1 & 4 & -1 \\ 1 & 1 & 1 \\ 0 & 2 & 3 \end{bmatrix}$ ○ $\begin{bmatrix} 3 & -1 & -1 & 50 \\ -1 & 4 & -1 & 100 \\ 1 & 1 & 1 & 120 \\ 0 & 2 & 3 & 350 \end{bmatrix}$ ○ $\begin{bmatrix} 3 & -1 & -1 & 50 \\ -1 & 4 & -1 & 100 \\ 1 & 1 & 1 & 120 \\ 2 & 3 & 0 & 350 \end{bmatrix}$ ○ $\begin{bmatrix} 3 & -1 & -1 \\ 4 & -1 & -1 \\ 1 & 1 & 1 \\ 0 & 2 & 3 \end{bmatrix}$

(b) (2 pt) Using MATLAB syntax and *one line of code*, define matrix $\mathbf{M}$.

```
M = [3 -1 -1; -1 4 -1; 1 1 1; 0 2 3]
% OR [3, -1, -1; -1, 4, -1; 1, 1, 1; 0, 2, 3]
```

(c) (1 pt) Using MATLAB syntax and *one line of code*, define vector $\mathbf{y}$.

```
y = [50; 100; 120; 350] % OR [50 100 120 350]'
```

(d) (2 pt) Which of the following statements is True for the above system of linear equations?

○ The system of linear equations has 4 unknowns

○ Matrix $\mathbf{M}$ is singular because `det(M)` is equal to 0

○ Using `det(M)` will allow us to know if the system has a unique solution or not

● None of the options

○ Matrix $\mathbf{M}$ is invertible because `det(M)` is not equal to 0

○ Using `det(M)` will allow us to know if the system has a unique, infinitely many, or no solution

(e) (2 pt) Based only on the following results, which of the following is True regarding the system of linear equations?

```
>> rank(M)
ans =
     3

>> rank([M, y])
ans =
     4
```

- ○ There is a unique solution
- ○ None of the options
- ● There is no solution
- ○ There are infinite solutions
- ○ There are two solutions
- ○ There are three solutions

(f) (3 pt) Regardless of your answer to (e), which of the following MATLAB expressions can be used to solve the given system of linear equations: $\mathbf{MT} = \mathbf{y}$? Select ALL that apply.

☐ `y * pinv(M)`	☐ `inv(y' * y) * y' * M`	☐ `y \ M`
■ `inv(M' * M) * M' * y`	■ `pinv(M) * y`	■ `M \ y`
☐ `M .\ y`	☐ `y / M`	☐ `M / y`
☐ `inv(M) * y`	☐ `pinv(y) * M`	☐ `pinv(M) .* y`

(g) (2 pt) Which of the following best describes the result returned by MATLAB after using one of the commands to solve the given system of equations, $\mathbf{MT} = \mathbf{y}$?

- ○ An infinite number of exact solutions
- ○ The exact solution that satisfies all equations
- ○ The least-squares solution that minimizes the MATLAB expression `norm(T * M - y)`
- ● The least-squares solution that minimizes the MATLAB expression `norm(M * T - y)`
- ○ The least-squares solution that maximizes the MATLAB expression `norm(T * M - y)`
- ○ The least-squares solution that maximizes the MATLAB expression `norm(M * T - y)`
- ○ The command does not return any result
- ○ A random solution
- ○ The command produces an error since the system has no solution

(h) (1 pt) Suppose that the third equation ($T_1+T_2+T_3 = 120$) is incorrect and should be excluded. Assume `M` and `y` are already defined (parts b and c). Write one MATLAB command to remove the third equation from `M` without redefining the entire matrix and while using `y`. Any solution that redefines the entire matrix or does not use `y` to exclude the third equation will receive 0.

```
M = M(y ~= 120, :);
```

Question 5 (10 points)

(8 pt) Each Python code block attempts to compute the average of numbers in a list. For each block, select what will be displayed after execution. Note that `Error` means the code raises an error. Select ALL that apply. Note that the built-in functions `sum(A)` and `len(A)` return the sum of all elements and the number of elements in list `A`, respectively.

(a)
```
print("start")
numbers = [1, 2, 3]
avg = sum(numbers) / len(numbers)
print(avg
print("end")
```

- ☐ 1
- ☐ 3
- ☐ end
- ☐ 2
- ■ Error
- ☐ start

(b)
```
print("start")
numbers = [1, 2, 3]
avg = sum(numbers) / len(numbers)
try:
    print(Avg)
except:
    print(1)
print("end")
```

- ■ start
- ☐ 2
- ☐ Error
- ■ 1
- ■ end
- ☐ 3

(c)
```
print("start")
numbers = [1, 2, 3]
avg = sum(numbers) / len(numbers)
print(Avg)
print("end")
```

- ■ Error
- ■ start
- ☐ 1
- ☐ 3
- ☐ end
- ☐ 2

(d)
```
print("start")
numbers = [1, 2, 3]
avg = sum(numbers) / len(numbers)
try:
    print(Avg)
except ZeroDivisionError:
    print(1)
print("end")
```

- ■ Error
- ☐ end
- ☐ 3
- ■ start
- ☐ 1
- ☐ 2

(2 pt) Similarly, this Python code block attempts to compute the average of numbers in a list. Which of the following statements is(are) True? Select ALL that apply.

(e)
```
print("start")
numbers = [1, 2, 3]
avg = sum(numbers) + len(numbers)
print(avg)
print("end")
```

- ■ All `print()` statements execute
- ☐ No `print()` statements execute
- ☐ An error will be raised
- ☐ There is a round-off error
- ■ There is a logical error
- ☐ Only some `print()` statements execute

Keep going — you're doing great!

Why do programmers prefer dark mode on their laptop screen?

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Because light attracts bugs :)

Question 6 (10 points)

You need to write a Python function `int2bin(n)` that takes a nonzero base-10 integer `n` (strictly negative or strictly positive but `n` cannot be 0) and returns a string representing the signed binary integer form of `n`. Refer to the examples below. Anything written on this page will NOT be graded.

Hint: We discussed in lecture how to convert a base-10 integer to binary, and also did several examples. The same method is applied here.

You may only use the built-in `str` function, which converts various data types into a string. No other functions or methods are allowed, built-in or otherwise.

```
def int2bin(n):
    """Convert a nonzero base-10 integer to signed binary integer.
    >>> int2bin(1)
    '01'
    >>> int2bin(-2)
    '110'
    >>> int2bin(5)
    '0101'
    >>> int2bin(-6)
    '1110'
    """
    # Determine the sign bit
    sign = ______(a)______
    # Work with the absolute value
    n = abs(n)
    # Initialize an empty string
    binary = ''
    ______(b)______:
        binary += ______(c)______
        n = ______(d)______
    return ______(e)______
```

You will implement the function `int2bin` by filling in the blanks. You may only write one statement on each line. You may not use ; to place two statements on the same line. You may only use the built-in `str` function. No other functions or methods are allowed, built-in or otherwise. You may not modify the level of indentation of any line.

(a) (2 pt) Which option correctly fills blank (a)?

- ○ `'-1' if n < 0 else '1'`
- ○ `'1' if n > 0 else '0'`
- ○ `'+' if n < 0 else '-'`
- ○ `str(n > 0)`
- ○ `'-' if n < 0 else '+'`
- ○ `str(n < 0)`
- ○ `'0' if n < 0 else '1'`
- ● `'1' if n < 0 else '0'`

(b) (3 pt) Fill in blank (b).

```
while n > 0: # OR while n != 0:
```

(c) (2 pt) Fill in blank (c).

```
binary += str(n % 2)
```

(d) (1 pt) Fill in blank (d).

```
n = n // 2
```

(e) (2 pt) Fill in blank (e).

```
return sign + binary[::-1]
```

Question 7 (15 points)

You need to write a MATLAB function `streak(s, lim)` which takes a vector `s` representing the speed of a car recorded every minute and a scalar `lim` representing the speed limit. The function should find the longest streak. A streak is a run of one or more consecutive speeds that are greater than or equal to the limit. The function should return: (1) `Start`, which is the first index of the longest streak, and (2) `Len`, which is the length of that streak. If there are multiple streaks of the same length, `Start` should be the beginning index of the first streak. If the car never exceeds the limit, the function should return `0` for both outputs.

Test Cases:

```
>> s = [25, 26, 20, 27];
>> [Start, Len] = streak(s, 25)
Start = 1
Len = 2

>> s = [25, 26, 20, 27, 28];
>> [Start, Len] = streak(s, 25)
Start = 1
Len = 2
```

```
>> s = [25, 26, 20, 27, 28, 25];
>> [Start, Len] = streak(s, 25)
Start = 4
Len = 3

>> s = [24, 20, 23];
>> [Start, Len] = streak(s, 25)
Start = 0
Len = 0
```

```
function [Start, Len] = streak(s, lim)

    % Initialize outputs
    Start = 0; Len = 0;

    % Initialize start index and length for current streak
    currentStart = 0; currentLen = 0;

    for ______(a)______

        if ______(b)______

            if ______(c)______

                currentStart = i;
            end
            currentLen = currentLen + 1;
        else
            ______(d)______;
        end

        if ______(e)______

            Len = ______(f1)______;

            Start = ______(f2)______;

        end
    end
end
```

You will implement the function by filling in the blanks. You may only write one statement on each line. You may not use ; or , to place two statements on the same line. You may not use ANY functions, unless given as multiple choices. You should use MATLAB syntax.

(a) (2 pt) Which option correctly fills blank (a)?

- ○ `i = s`
- ○ `i in s`
- ○ `i in length(s)`
- ● `i = 1:length(s)`
- ○ `i = range(1, len(s))`
- ○ `i = range(len(s))`
- ○ `i = length(s)`
- ○ `i in 1:length(s)`

(b) (2 pt) Fill in blank (b).

```
if s(i) >= lim
```

(c) (1 pt) Which option correctly fills blank (c)?

- ○ `currentLen = 0`
- ○ `Len = 0`
- ○ `currentStart = 0`
- ○ `currentLen = 1`
- ○ `currentLen == 1`
- ○ `Len == 0`
- ○ `s > lim`
- ○ `Len == 1`
- ○ `s < lim`
- ○ `Len = 1`
- ● `currentLen == 0`
- ○ `currentStart == 0`

(d) (2 pt) Fill in blank (d).

```
currentLen = 0
```

(e) (2 pt) Fill in blank (e).

```
if currentLen > Len
```

(f) (2 pt) Fill in blanks (f1) and (f2).

```
Len = currentLen
```

```
Start = currentStart
```

(g) (2 pt) Next, assume you called your function and obtained `Start` and `Len`. You now want to calculate `Dis`, the total distance traveled (in km) during the longest streak. Note that distance is speed multiplied by time and `s` represents the car's speed recorded every minute in km/hr. Assume the car's speed remains constant during each one-minute interval. Write a single line of code to calculate `Dis`. You may not use ; to place two statements on the same line. You may only use the built-in function `sum(A)`, which returns the sum of the elements of array `A`.

```
% Assume s and lim are already defined
[Start, Len] = streak(s, lim);

Dis = sum(s(Start:Start+Len-1) / 60)
```

(h) (2 pt) Finally, you want to define a MATLAB function `c(s, lim)` that returns the number of times the car's speed transitions from being greater than or equal to the limit to being less than the limit. The inputs are the same as before, where `s` is a vector of the car's speed and `lim` is the speed limit, both in km/hr. Your full answer should be a single line of code. You may not use ; to place two statements on the same line. You may only use the built-in function `sum(A)`, which returns the sum of the elements of array `A`. No partial credit will be awarded for incorrect solutions.

Test Cases:

```
>> c([26, 25, 20, 27], 25)
ans = 1

>> c([25, 26, 20, 27, 24], 25)
ans = 2
```

```
>> c([25, 19, 27, 23, 25, 22], 25)
ans = 3

>> c([24, 20, 23], 25)
ans = 0
```

```
c = @(s, lim) sum(s(1:end-1) >= lim & s(2:end) < lim);
```